

F1 Performance Analysis

A Data-Driven Look at Formula 1 (1950-2024)

About the Data

This dataset is from Kaggle (accessible here:

<https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>), originally compiled from the Ergast Developer API. It covers every Formula 1 season from 1950 to 2024 across 14 CSV files. All 14 tables were loaded into Google BigQuery and joined into a single master view called `f1_master` for analysis. Each record represents one driver's result in one race, and includes:

- `year / round / race_name / race_date`: When and where the race took place
- `circuit_name / country`: The track and country it was held in
- `forename / surname / driver_nationality`: Driver name and nationality
- `constructor_name / constructor_nationality`: The team the driver raced for
- `grid`: Starting position on the grid
- `position / positionOrder`: Finishing position (position is null for non-finishers; positionOrder always has a value)
- `points`: Championship points earned in the race
- `laps`: Number of laps completed
- `finish_time`: Total race time for classified finishers
- `fastestLapTime / fastestLapSpeed`: The driver's fastest lap in the race
- `status`: How the race ended (Finished, Retired, Accident, etc.)
- `quali_position / q1 / q2 / q3`: Qualifying session times and final grid position
- `total_pit_stops / avg_pit_duration_ms / first_pit_lap`: Pit stop summary per driver per race
- `avg_lap_ms / best_lap_ms / worst_lap_ms / lap_time_stddev`: Lap time statistics per driver per race

Several columns in the raw data use `\N` as a placeholder for missing values, a quirk of the Ergast API export format. These were replaced with proper nulls directly in the BigQuery view using `NULLIF`. Qualifying, pit stop, and lap time data is also limited before 2011, so analysis using those fields focuses on 2011-2024.

Problem Statement

Formula 1 produces a huge amount of data every race weekend, but most of it stays buried in timing sheets and race reports. The goal of this project is to find patterns across 70+ years of racing history: which teams have dominated, whether starting position actually determines the winner, how pit stop strategy affects results, what separates consistent drivers, and how much faster cars have gotten at the circuits that show up year after year.

Research Questions

- Which constructors have dominated F1 over the past two decades, and how has competitive balance shifted over time?
- How does qualifying grid position relate to final race finishing position? Does starting near the front actually determine who wins?
- What is the relationship between pit stop strategy (number of stops, timing, and duration) and where a driver finishes?
- Which drivers show the most consistent lap times across a season, and does that consistency track with championship results?
- How have average lap times at major circuits changed over the years?
- What do DNF and reliability patterns look like across constructors?

Data Preparation

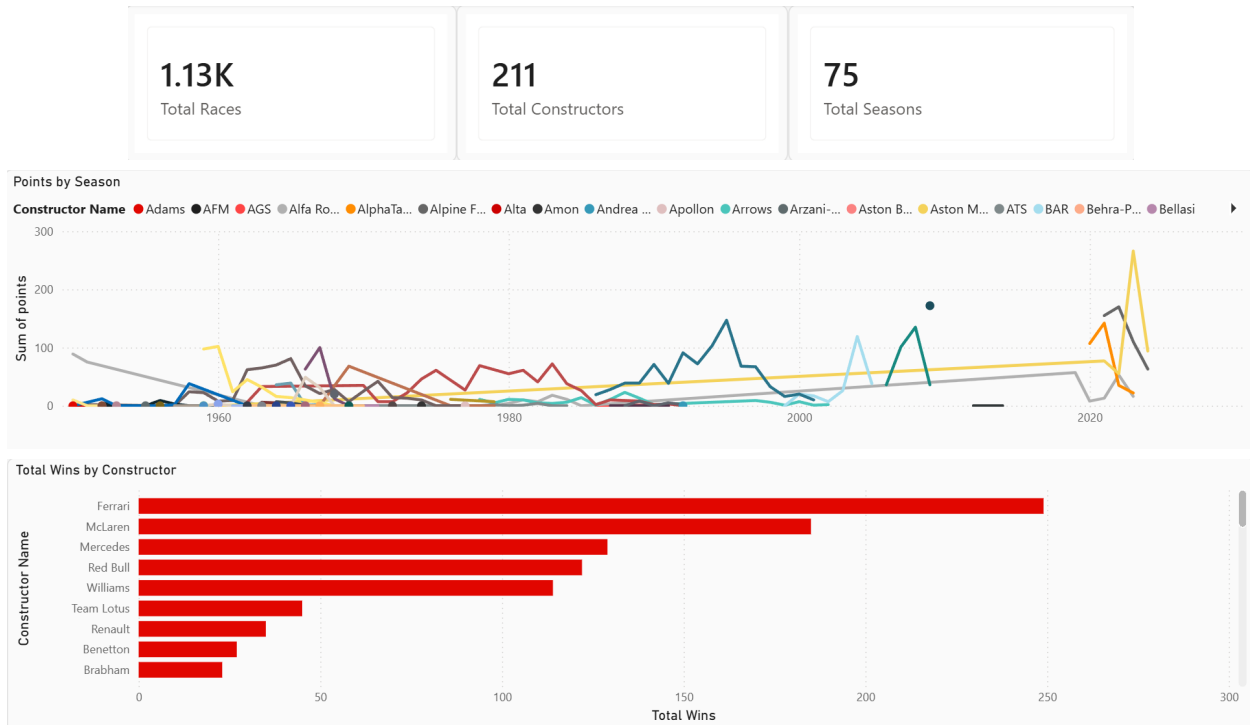
The main preparation work happened in two places: the BigQuery view and the EDA notebook.

In BigQuery, the `f1_master` view was built to join all 14 source tables into one flat table. `results` was the base, with INNER JOINS to `races`, `circuits`, `drivers`, `constructors`, and `status` since every result has to have a match in those. `qualifying`, `pit_stops`, and `lap_times` used LEFT JOINS since older races often have no data for those columns. Pit stop and lap time data were pre-aggregated into per-driver per-race summaries using CTEs so the view stayed at one row per driver per race. The `\N` placeholders were cleaned up with `NULLIF` across all affected columns.

In the EDA notebook, a few columns were added for analysis. A `positions_gained` field was calculated as qualifying position minus finish position to measure how much a driver moved through the field. A `result_category` column classified each result as Finished, DNF, or DNS. Lap time consistency was measured using coefficient of variation (standard deviation divided by mean lap time) to make it comparable across different circuits and seasons.

Analysis

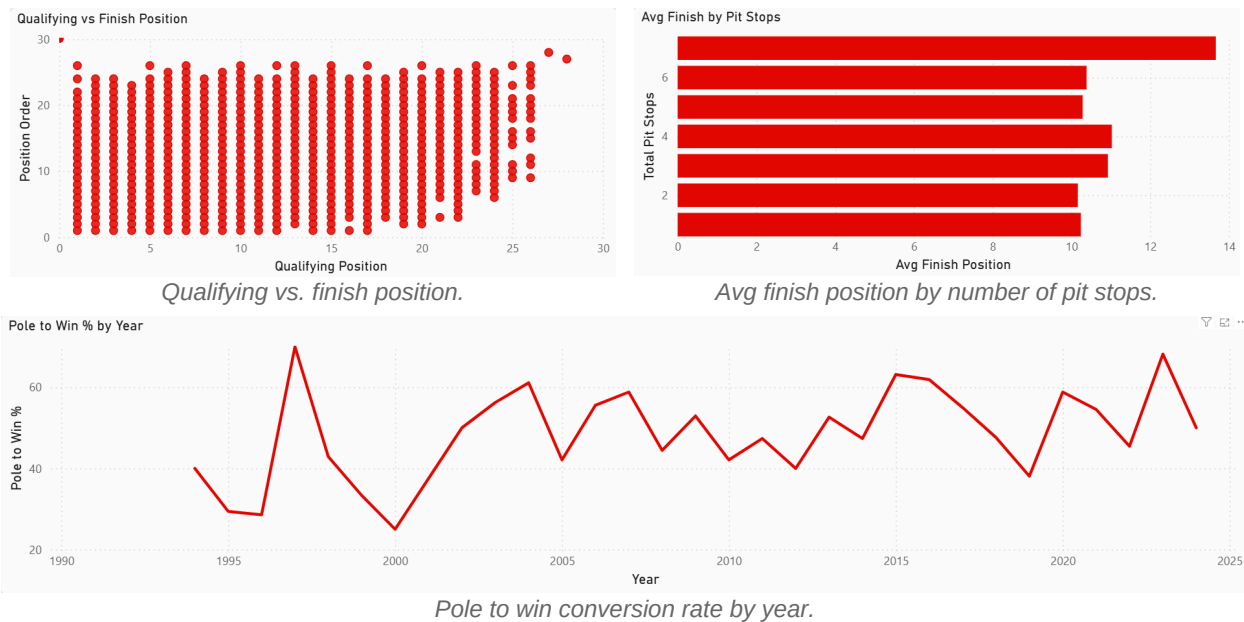
Which constructors have dominated F1, and how has competitive balance shifted?



Constructor points by season (top) and total wins by constructor (bottom).

Constructor points were aggregated by season and analyzed using the Herfindahl-Hirschman Index (HHI), a measure of market concentration adapted here to show how evenly points are spread across teams each season. A higher HHI means one or two teams are dominating; a lower number means the field is more competitive. Three dominant eras stand out: Red Bull 2011-2013, Mercedes 2014-2021, and Red Bull again in 2022-2023. Ferrari leads in total wins across the full dataset going back to 1950, but that reflects their longevity more than recent dominance. 2012 had the lowest HHI in the dataset, lining up with one of the most competitive modern seasons, seven different winners in the first seven rounds.

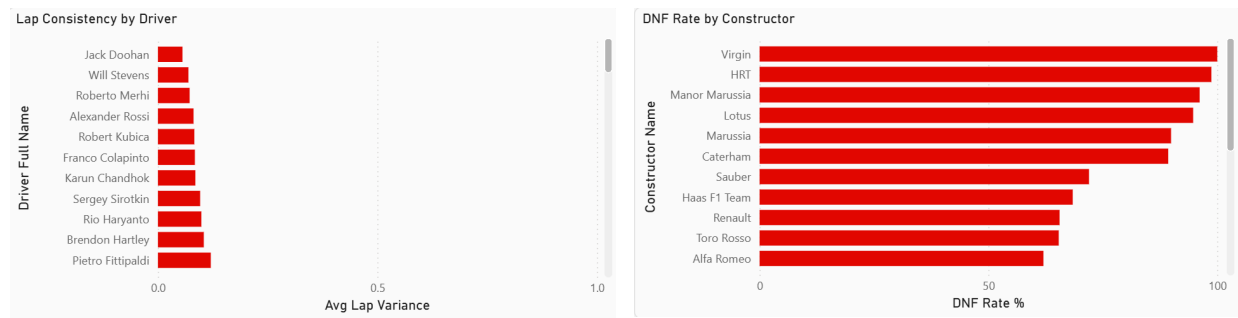
How does qualifying position relate to finishing position, and does pit stop strategy matter?



Pearson correlation between qualifying position and finishing position came out at $r = 0.65$, which is a real but imperfect relationship. Starting near the front matters, but it is far from a guarantee. Pole-to-win conversion rate peaks during dominant car eras and drops in competitive seasons like 2012 and 2021 where the field is closer together.

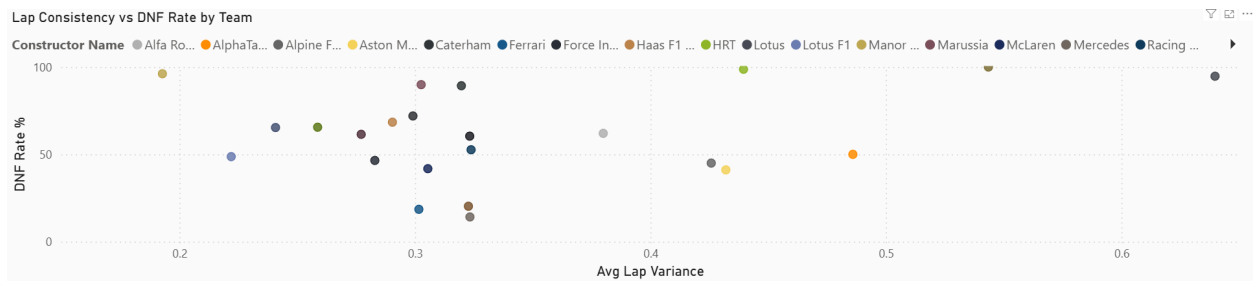
On pit stop strategy, one-stop races produce the best average finishing positions among classified finishers. Four or more pit stops almost always reflects reactive strategy forced by incidents or tire failures, and it correlates with worse finishes. The data supports keeping strategy simple when possible.

Which drivers are the most consistent, and what do DNF patterns look like by team?



Lap consistency by driver (lower variance = more consistent).

DNF rate by constructor.



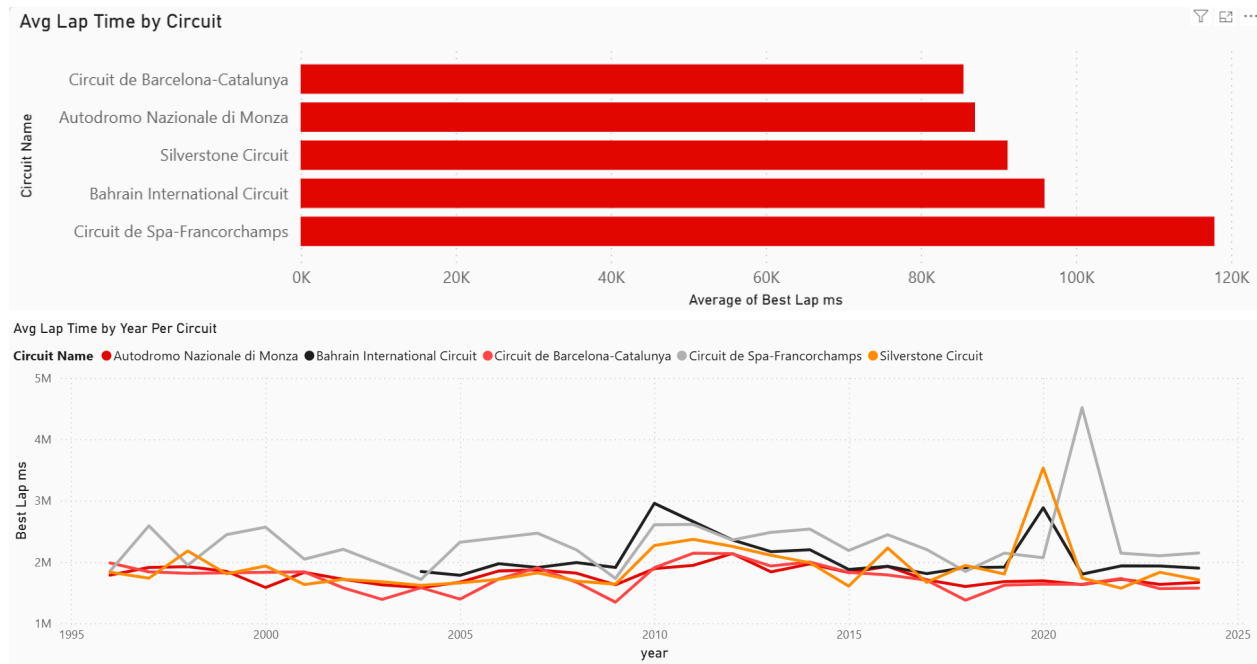
Lap variance vs. DNF rate by team.

Lap time consistency was measured using coefficient of variation (CV), which scales the standard deviation by the mean so it is comparable across different circuits and seasons. There is a negative correlation between lap CV and season points, more consistent drivers tend to score more. The consistency chart shows the tightest lap timers across the 2011-2024 period.

On reliability, smaller teams DNF more often. Virgin, HRT, and Manor Marussia sit at the top of the DNF rate chart, reflecting underfunded backmarker operations running at the limit. The scatter plot shows teams with higher lap variance also tend to have higher DNF rates, though it is not a perfect relationship.

How have lap times changed at major circuits over the years?

Circuit Name	Year	Best Lap ms
Bahrain International Circuit	2020	55404



Fastest recorded lap (top), avg lap time by circuit (middle), avg lap time by year per circuit (bottom).

The fastest recorded lap in the dataset was set at Bahrain International Circuit in 2020 at 55,404ms (just over 55 seconds). Across the five major circuits tracked, Circuit de Spa-Francorchamps has the highest average lap time due to its length, while Barcelona and Monza tend to be faster. Lap times dropped noticeably starting in 2017 after the wider car regulations, and the 2022 ground effect reset briefly slowed things down before teams recovered pace by 2023-2024. The large spikes visible around 2020-2021 in the line chart reflect some circuits only appearing in the calendar for one or two years during the COVID-era schedule, skewing their averages.

Conclusion

The analysis confirmed a few things that F1 fans already suspect but rarely see backed up with data. Regulation changes drive competitiveness more than anything else. The most dominant eras line up almost perfectly with regulation periods that gave one team a sustained technical advantage. The most competitive seasons happen during transitions when no one has figured it out yet.

Qualifying position matters but does not determine the winner. The 0.65 correlation is meaningful but leaves plenty of room for strategy, reliability, and racecraft to flip the result. The pole-to-win chart makes this clearest, it spikes when one team has a dominant car and flattens out when the field is close.

On pit stops, the data supports keeping it simple. One-stop races produce better finishes on average, and drivers who end up doing four or more stops are almost always reacting to something going wrong.

A few areas worth exploring further: whether the budget cap shows up as a measurable shift in the HHI trend, whether street circuits vs. high speed tracks change how much qualifying position matters, and whether a simple model using qualifying position, team, and pit stop count can predict finishing position better than just using qualifying alone.